Causal Inference & Synthetic Control Engine for Algorithmic Pricing

Final Project Report | Mohd Sahil

Project overview. This project demonstrates a workflow for exploring a simulated price increase in Delhi across five markets, combining EDA, Synthetic Control, placebo checks, DML, sensitivity analysis, and a Streamlit dashboard.

Research Question & Dataset

The simulated intervention raises Delhi's price from ■1,000 to ■1,100 beginning 1 May 2025. The generated dataset contains five markets with 180 daily observations per market (900 market-day rows), using a synthetic demand process with seasonality, shared shocks, advertising spend, and random noise.

Methods

- **EDA:** Reviews daily revenue and market-level patterns.
- **Synthetic Control:** Creates a weighted control-market comparison using pre-intervention revenue.
- **Placebo checks:** Repeats the approach with other markets as placebo-treated units.
- **DML:** Uses cross-fitting and residualization to estimate a treatment coefficient adjusted for observed covariates.
- **Sensitivity analysis:** Compares estimates across feature specifications.

Selected Results From Prior Runs

Analysis	Previously reported result	Interpretation
Synthetic Control gap	≈ ■10,348.60/day	Average post-period Delhi minus synthetic revenue gap.
Pre-treatment RMSE	≈ ■4,980.62	Fit error; inspect the full pre-period plot too.
DML estimate	≈ ■10,654.48	Model-based estimated treatment effect.
Approx. 95% CI	■7,816.66–■13,492.30	Approximate interval from previous implementation.
Sensitivity	≈ ■10,654–■14,659	Effect changes with model specification.

These figures are from earlier analysis runs. Re-run the final scripts and align report values with the active dataset and dashboard before publishing.

Interpretation

The previously reported Synthetic Control and DML estimates are positive, but sensitivity estimates vary and placebo results are mixed. This is best presented as an exploratory methods demonstration rather than a definitive causal finding.

Limitations

- The dataset is synthetic and does not represent real business outcomes.
- Only Delhi is treated; a Delhi-specific post-period shock could be confused with a pricing effect.
- Synthetic Control requires a credible pre-treatment fit and comparable controls; some placebo fits were weak.
- There are only four potential donor markets, limiting placebo inference.
- The clustered DML confidence interval is approximate; many dates do not compensate for having only one treated market.
- DML adjusts for measured covariates but cannot automatically eliminate unobserved confounding.

Final Reproducibility Checklist

- Regenerate the dataset and confirm which CSV is active.
- Run every analysis script and save final plots/results.

- Compare script outputs with dashboard outputs.
- Confirm requirements files include all required libraries.
- Update the README and report with final rerun values.
- Clearly label the project as a synthetic-data demonstration.

Conclusion

The project brings together data generation, visualization, counterfactual comparison, placebo checks, machine-learning-based adjustment, sensitivity analysis, and interactive presentation. Its portfolio value is the complete workflow and transparent discussion of assumptions and uncertainty. Stronger causal claims would require more treated units, stronger design diagnostics, and validation on appropriate real-world data.